Data Management for Machine Learning Assignment

Short Demo Link: https://drive.google.com/file/d/1qIWfoYd7LYJFkJoAOQU6cmr7-TYUYkE8/view?usp=drive_link

Github repo: https://github.com/pandeylokesh124/dm4ml

Group 36 Team information:

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1. Problem Definition

RecoMart, a rising e-commerce startup, faces the challenge of low customer conversion rates and limited cross-selling success. While the platform generates significant traffic, users often struggle to find products relevant to their interests, leading to high bounce rates.

The objective of this project is to design and implement an **automated, modular data management pipeline** that powers a personalized product recommendation engine. By leveraging historical user behaviour and product metadata, the system will deliver "You Might Also Like" suggestions, thereby improving user engagement, increasing average order value, and driving overall sales conversions.

2. Project Objectives

The primary technical objectives for the data platform team are:

- **Automated Ingestion:** Establish a reliable mechanism to fetch user interactions and product metadata from disparate sources.
- **Data Integrity:** Implement automated validation to ensure the recommendation model learns from high-quality, "clean" data (e.g., valid 1–5 rating scales).
- **Feature Centralization:** Build a structured Feature Store to maintain versioned, reusable features like user activity frequency and item popularity.
- **Orchestrated Workflow:** Utilize Apache Airflow to automate the sequence of tasks from raw data collection to model training.
- **Reproducibility:** Implement data and model versioning (DVC/MLflow) to track experiments and ensure lineage.

3. Key Data Sources

The pipeline will ingest and process two primary categories of data:

1. **User Interactions (Behavioral Data):**
    - **Source:** interactions.csv (Simulated clickstream and purchase logs).
    - **Attributes:** user_id, item_id, rating (1–5 scale), and timestamp.
    - **Purpose:** To capture collaborative patterns (which users liked which products) (p. 1).
2. **Product Metadata (Catalog Data):**
    - **Source:** External REST API (Simulated JSON response).
    - **Attributes:** product_id, category, price, and stock_status.
    - **Purpose:** To enable content-based filtering and feature enrichment (e.g., normalizing prices).

4. Expected Pipeline Outcomes

Successful execution of the pipeline will yield the following deliverables:

- **Validated Data Lake:** A structured local filesystem (partitioned by timestamp) containing raw and cleaned datasets ready for Exploratory Data Analysis (EDA).
- **Engineered Feature Store:** A repository of transformed features (e.g., Average Rating per Item) that can be retrieved for both batch training and real-time inference.
- **Trained SVD Model:** A Collaborative Filtering model (Matrix Factorization) tracked via MLflow, including its parameters and run IDs.
- **Automated DAG:** A functional Airflow Directed Acyclic Graph (DAG) that visualizes and manages the end-to-end data flow.

5. Evaluation Metrics

To measure the success of the recommendation engine, the following metrics will be tracked:

- **Mean Absolute Error (MAE):** To measure the deviation between predicted and actual user ratings.
- **Precision@K:** The proportion of recommended items in the top-K set that are truly relevant to the user.
- **Recall@K:** The proportion of all relevant items that were successfully captured in the top-K recommendations

Data Profiling and Validation: data_quality_report.pdf file attached

Data Quality Report

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Summary Metrics

Input file: sample_interactions.csv

Raw row count: 299

Raw column count: 4

Cleaned row count: 292

Cleaned column count: 4

Duplicate rows removed: 7

Total missing values handled: 0

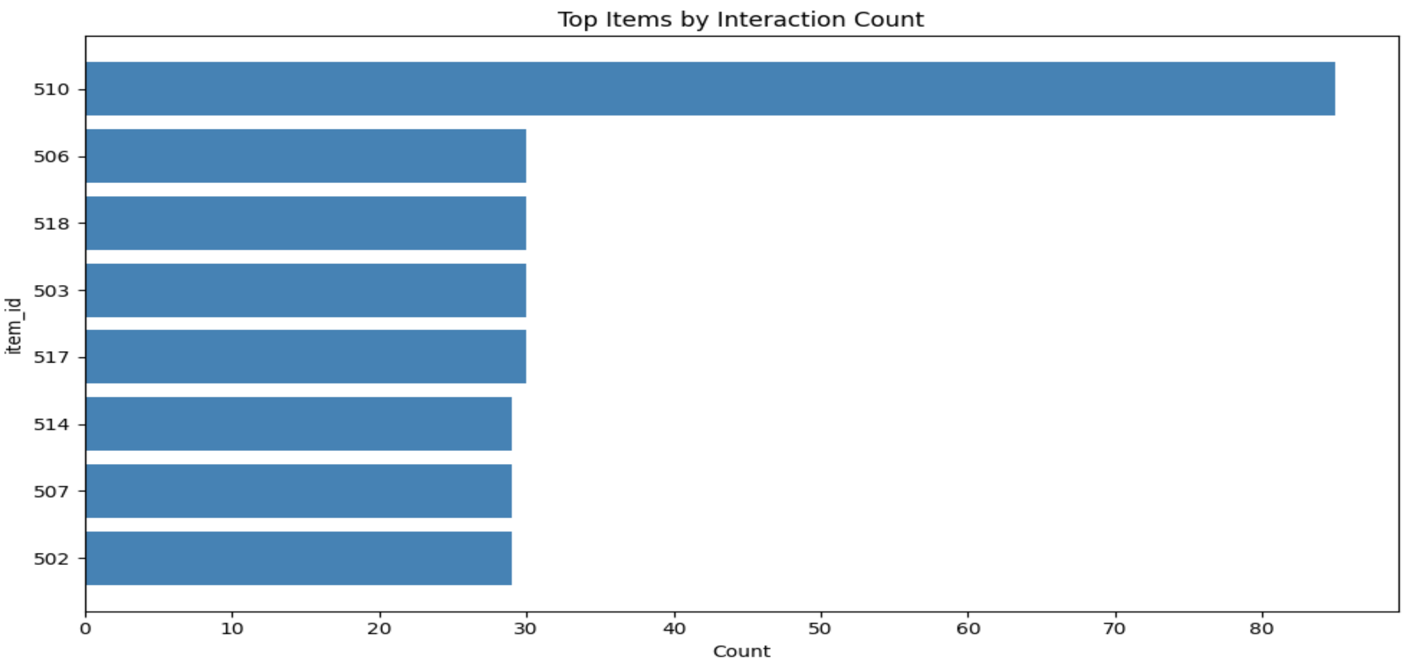
Issues & Validation Notes

- Extra columns detected: timestamp

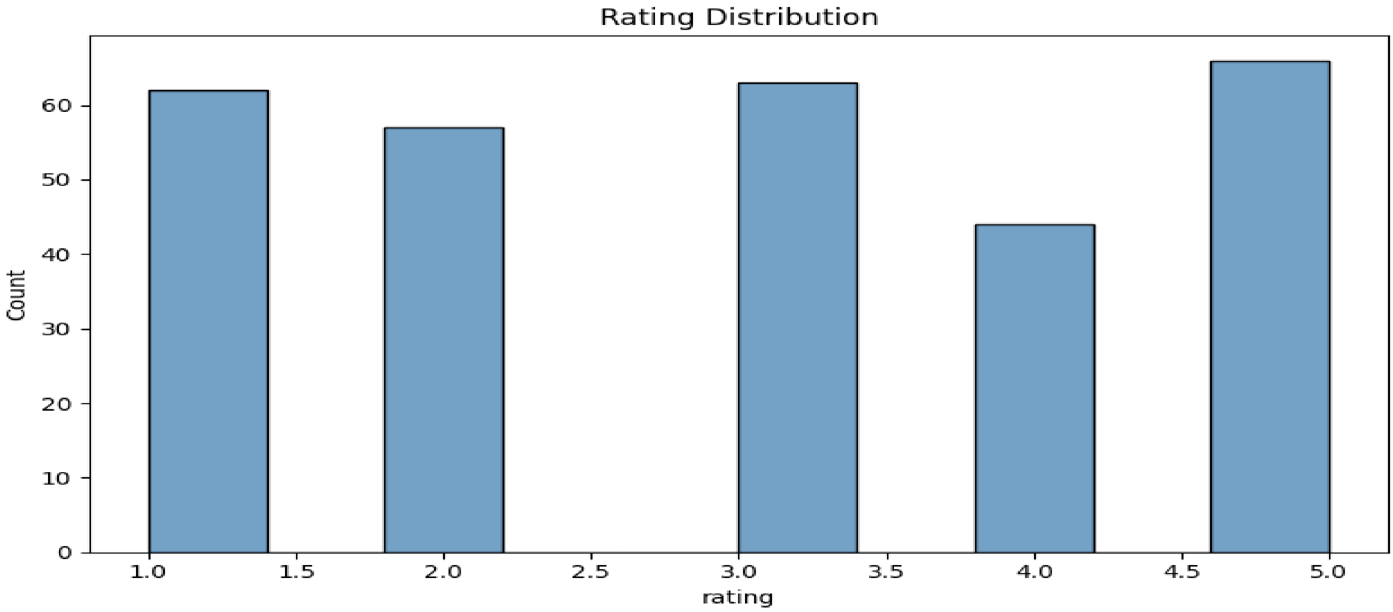
- Found 7 duplicate user-item rows.

Data Preparation: Summary plots screenshots

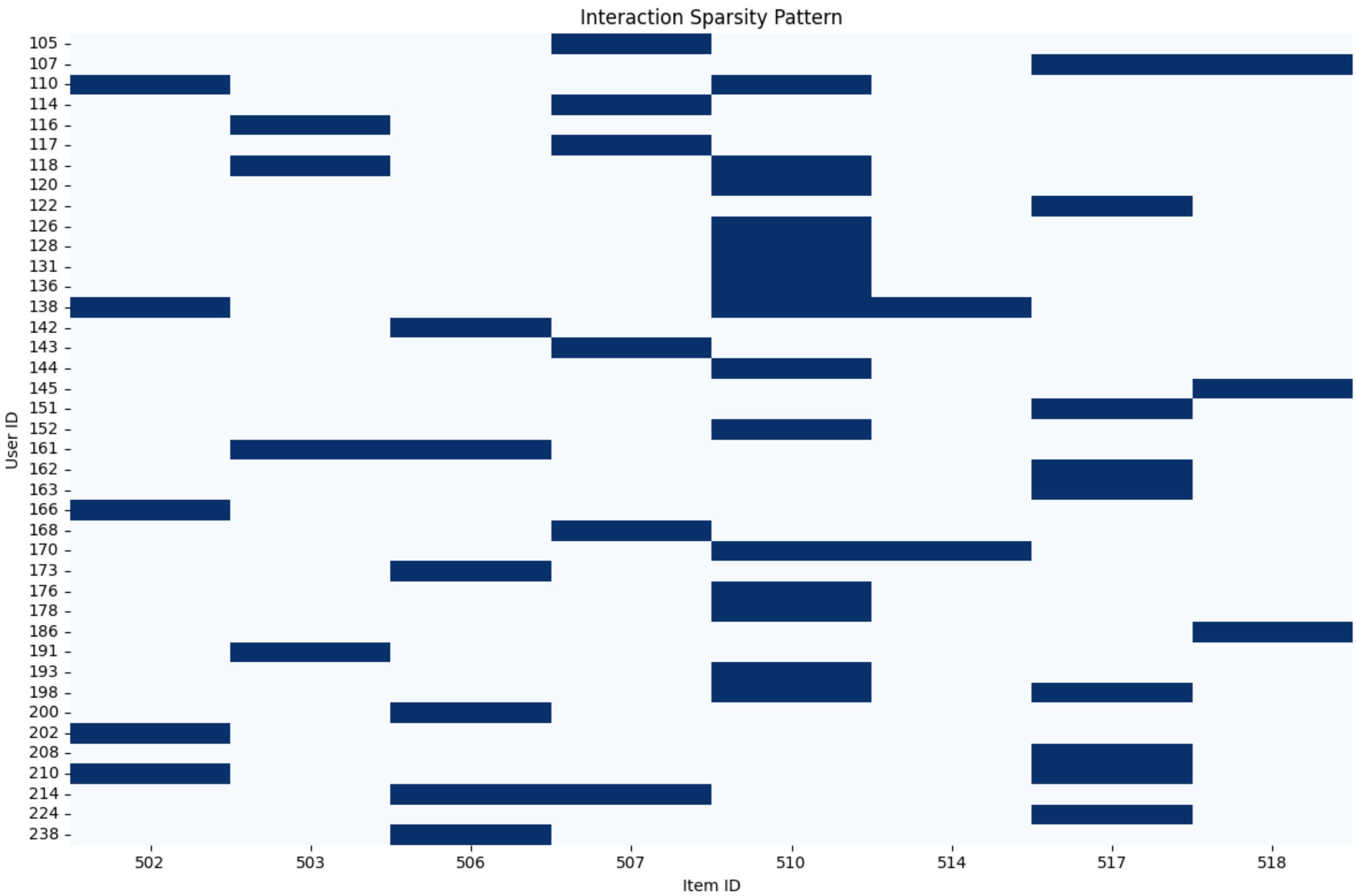
Item popularity



Rating distribution

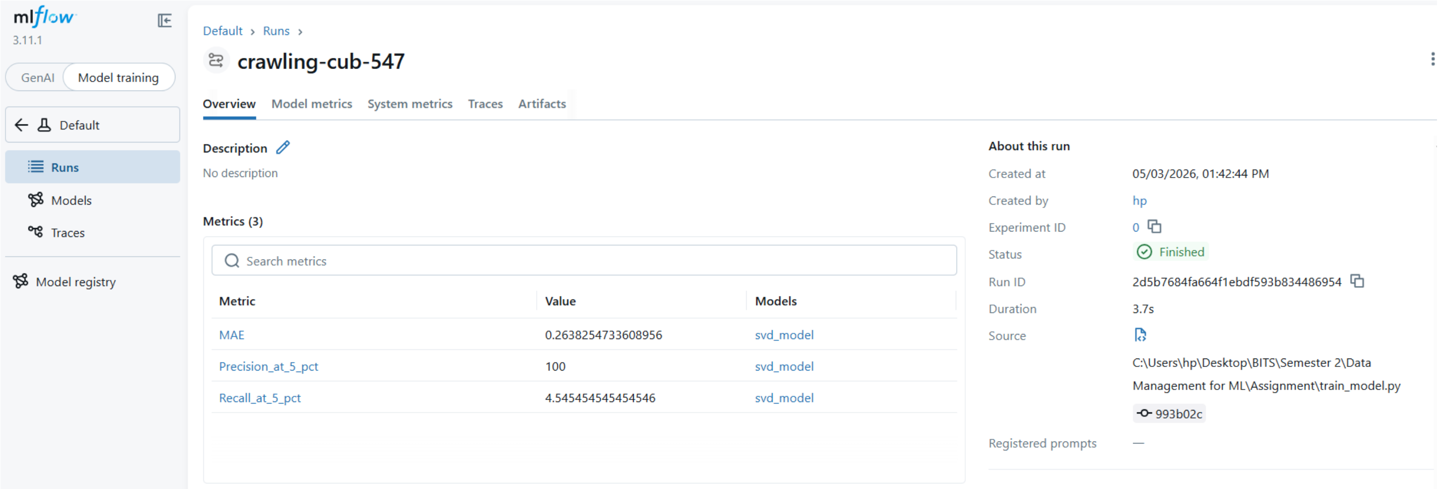


Interaction sparsity

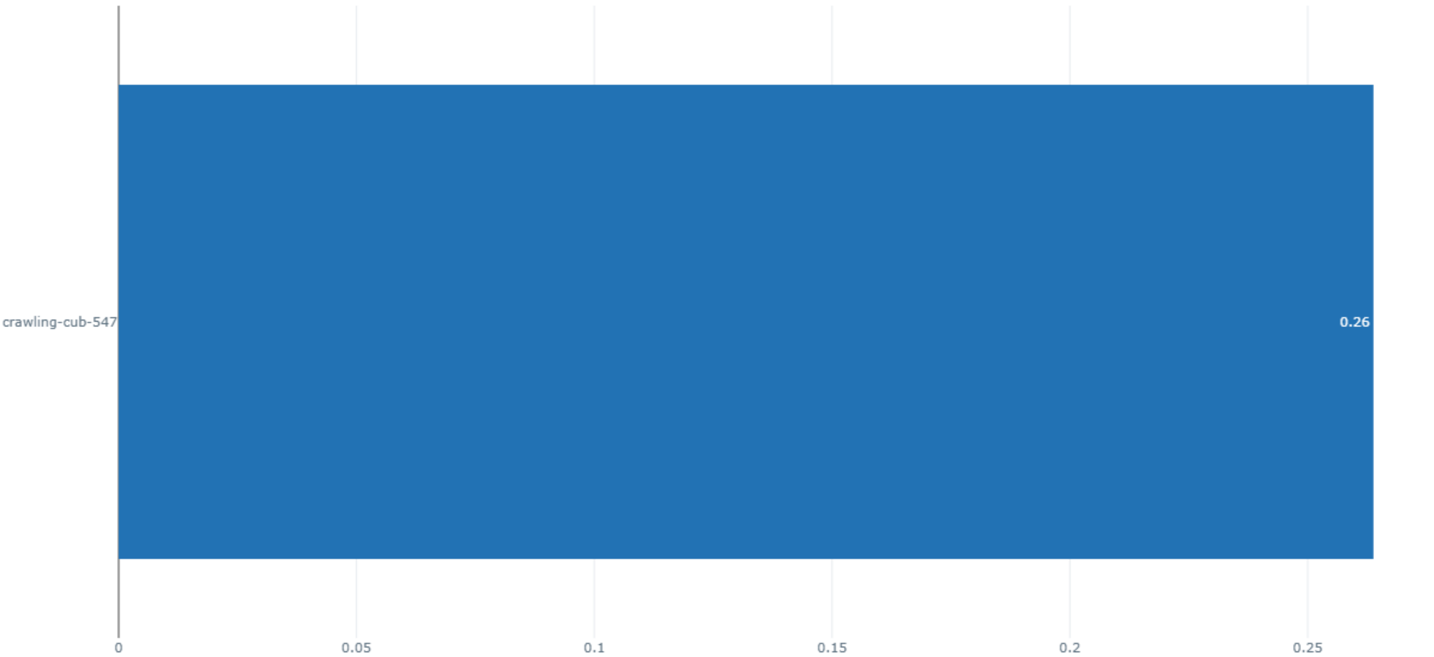


Model Training and Evaluation:

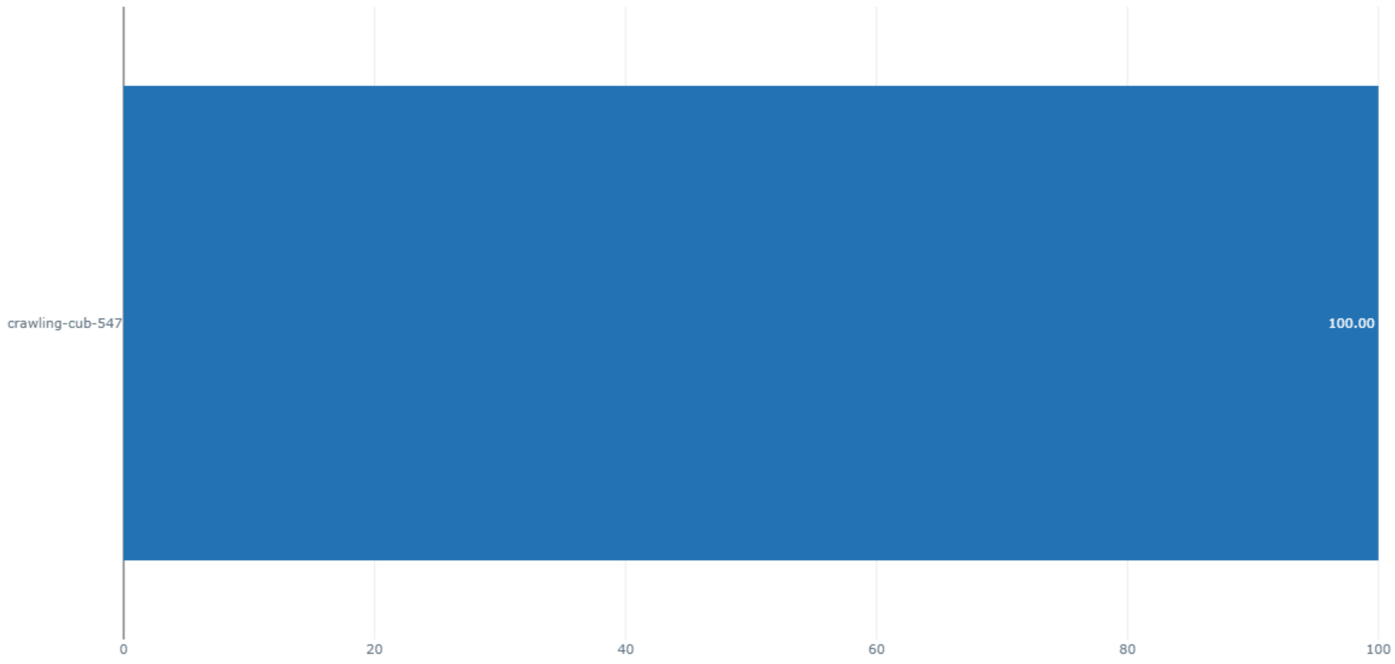
Model Performance Report from MLFlow UI



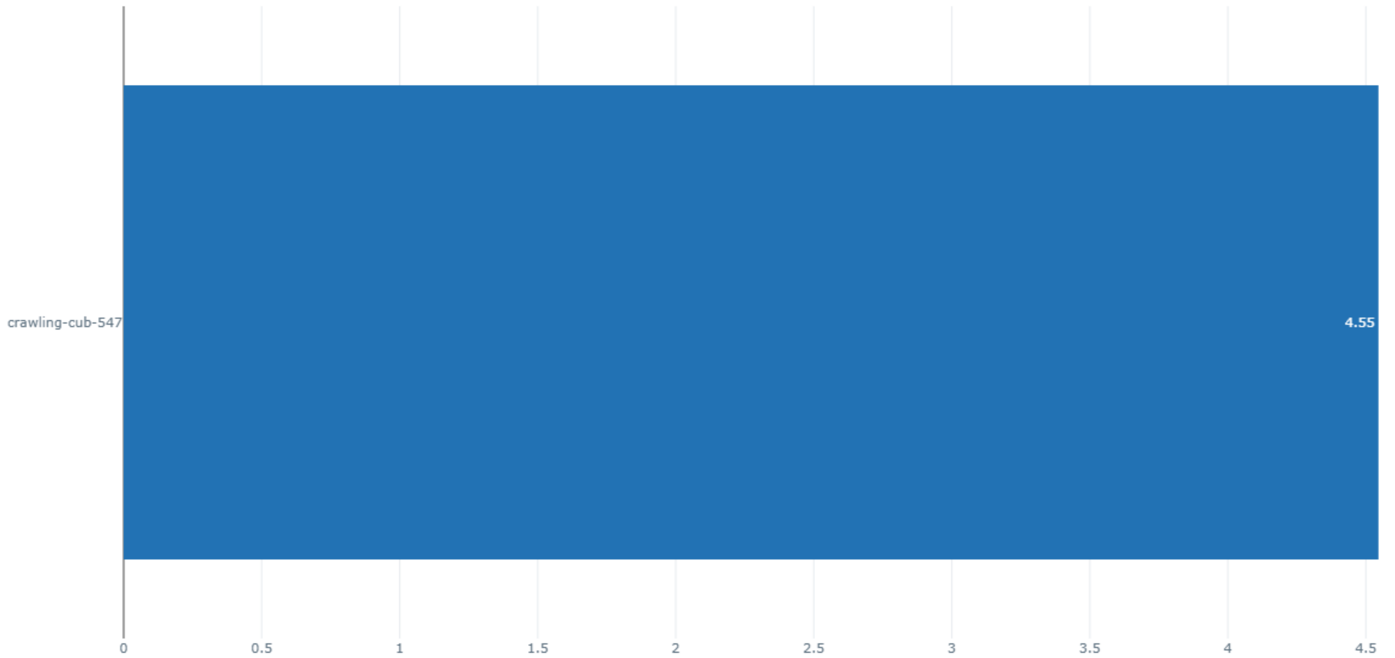
MAE



Precision@5



Recall@5



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